

UNIT- V: LASERS AND OPTICAL FIBERS

LASERS:

LASER: The word LASER stands for *Light Amplification by the Stimulated Emission of Radiation*. Laser is a source producing an intense, monochromatic, and highly parallel beam of coherent light.

Properties of Laser:

The extraordinary features of a laser beam are

1. High intensity - Laser beams are highly intense as a large number of photons are concentrated in a small region.
2. Monochromatic - Highly monochromatic radiation (i.e. single wavelength)
3. Coherence - Perfectly coherent as the emitted light waves has the same phase with one another.
4. High degree of directionality - Travels in a single direction as the photons are traveling along the optical axis of the system.

Basic Principles:

Interaction of an electromagnetic radiation with matter leads to transition of an atom or molecule from one energy state to another. If the transition is from lower state to higher state it absorbs the incident energy. If the transition is from higher state to lower state it emits a part of its energy.

If ΔE is the difference between the two energy levels,

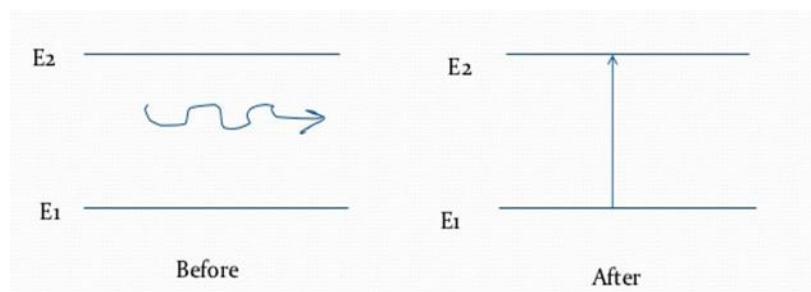
Then $\Delta E = (E_2 - E_1)$ Joule

According to Max Planck, $\Delta E = (E_2 - E_1) = h\nu = hc/\lambda$

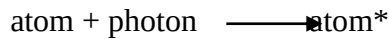


There are three possible ways through which radiation can interact with matter.

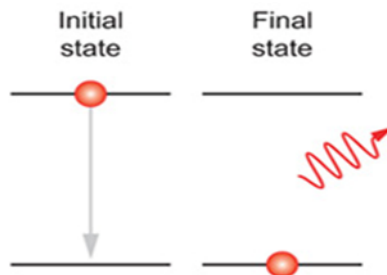
1) Induced Absorption: Induced absorption is the absorption of an incident photon by a system, as a result of which the system is elevated from a lower energy state to a higher state, where the energy of the incident photon is equal to the energy difference of the two energy states.



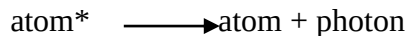
Let E_1 and E_2 be two energy levels in the energy level scheme of an atom in which E_1 is lower energy state. Let a photon having energy, $h\nu = \Delta E = E_2 - E_1$ be incident on the atom which is in the lower energy state. The atom absorbs the photon. Hence its energy becomes $E_1 + \Delta E = E_2$. Hence it is shown in the level E_2 . Then the atom is said to have made transition to the excited state. This can be represented as,



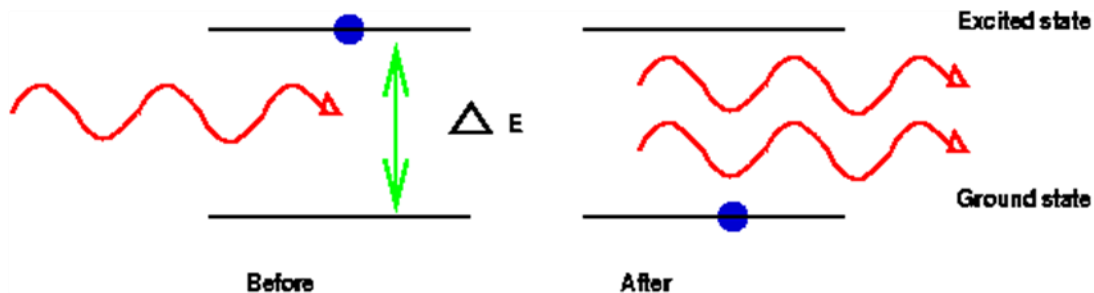
2) Spontaneous Emission: Spontaneous emission is the emission of a photon, when a system transits from a higher energy state to a lower energy state without the aid of any external agency.



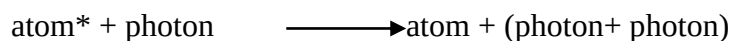
Consider an atom in the excited state. Because of universal tendency for any system to attain the least available energy state the atom emits a photon of energy $\Delta E = E_2 - E_1$. Then the energy of the atom becomes equal to $E_2 - \Delta E = E_1$, and is shown in level E_1 . Since the atom emits a photon, without the aid of external agency, it is called spontaneous emission. This process can be denoted as,



3) Stimulated Emission: Stimulated emission is the emission of a photon by a system, under the influence of an incident photon of right energy due to which the system transits from a higher energy state to a lower energy state. The photon thus emitted is called the stimulated photon (emitted photon) and will have the same phase, energy and direction of movement as that of the stimulating photon (incident photon).



Consider an atom in the excited state. Let a photon having an energy $\Delta E = E_2 - E_1$ interacts with atom by passing in its vicinity. The atom emits a photon and transits to the lower energy state. The two photons travel in exactly the same direction, and same energy. The waves associated with the two photons will have identical phase and thus they are coherent. The process can be represented as,



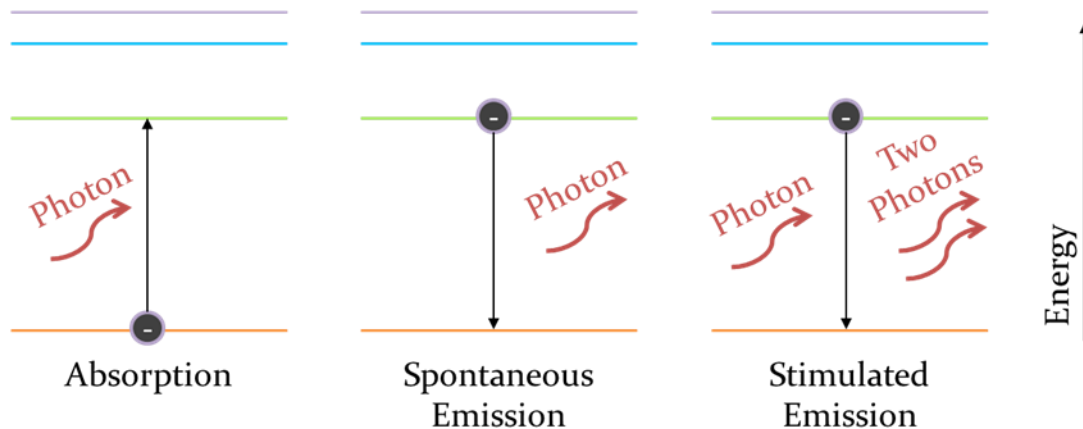
This kind of emission is responsible for laser action.

Einstein's Coefficients:

Einstein's coefficients give the probability associated with the absorption and emission processes.

Consider two energy states E_1 and E_2 of a system of atoms. Let N_1 and N_2 be the number density of atoms in the states 1 and 2 (number density is the number of atoms per unit volume).

Let a radiation of frequency ν be incident on the system. Let U_ν represents the energy density of frequency ν (energy incident/unit volume of the system is called energy density).



i) Case of Induced Absorption:

In case of induced absorption, an atom in the level E_1 can go to the level E_2 , when it absorbs a radiation of suitable frequency $\nu = (E_2 - E_1)/h$. The number of such absorptions per unit time per unit volume is called rate of absorption. The rate of absorption depends upon.

- 1) the number density of lower energy state N_1 and
- 2) the energy density U_ν

$\therefore$ Rate of absorption $\propto N_1 U_\nu$

or Rate of absorption = $B_{12}N_1 U_\nu$

where B_{12} is called *Einstein's coefficient of induced absorption*.

ii) Case of spontaneous Emission:

In the case of spontaneous emission an atom in the higher energy level E_2 undergoes transition to the lower energy level E_1 by emitting a photon. This transition is independent of the energy density of any frequency in the incident radiation. The rate of spontaneous emission (number of emissions/unit volume/unit time) is proportional to only the number density in the higher energy state N_2 .

$\therefore$ Rate of spontaneous emission $\propto N_2$

$\therefore$ Rate of spontaneous emission = $A_{21} N_2$

where A_{21} is called the *Einstein's coefficient of spontaneous emission*.

iii) Case of stimulated Emission:

In stimulated emission an external photon of appropriate frequency stimulates the downward transition of an atom in the excited state. The number of stimulated emission per unit time per unit volume, ie, rate of stimulated emission is proportional to

1) number density of higher energy state N_2 and

2) the energy density U_ν

∴ Rate of stimulated emission $\propto N_2 U_\nu$

∴ Rate of stimulated emission $= B_{21} N_2 U_\nu$

where B_{21} is called **Einstein's coefficient of stimulated emission**.

From **Boltzmann's law**, we know that the ratio of the population densities N_2 and N_1 in the excited state E_2 and ground state E_1 respectively at any instant at any temperature T is given by,

$$\frac{N_2}{N_1} = e^{-(E_2 - E_1)/kT} = e^{-h\nu/kT} = e^{-hc/\lambda kT}$$

This equation indicates that the population at energy level E_2 is much smaller than that at E_1 at equilibrium and most of the electrons are in the lower energy state.

Expression for Energy density:

At steady state, we can write

Rate of absorption = rate of spontaneous emission + rate of stimulated emission

ie. $B_{12} N_1 U_\nu = A_{21} N_2 + B_{21} N_2 U_\nu$

Or $U_\nu (B_{12} N_1 - B_{21} N_2) = A_{21} N_2$

∴ By dropping subscript $B_{12} = B_{21} = B$ and $A_{21} = A$

∴ At thermal equilibrium, $U_\nu = \frac{A}{B[e^{(h\nu/kT)} - 1]}$]

Requisites of a Laser System (Laser Components)

The three requisites of a laser system are:

1) An excitation source for pumping action (pumping mechanism): The act of exciting atoms from lower energy state to a higher energy state by supplying energy is called pumping. The excitation source provides the appropriate amount of energy for pumping the atoms. The energy input may be in the form of light energy, electrical energy or chemical etc. The different pumping methods are

➤ Optical Pumping: By using high intensity light or flash tubes for excitation.

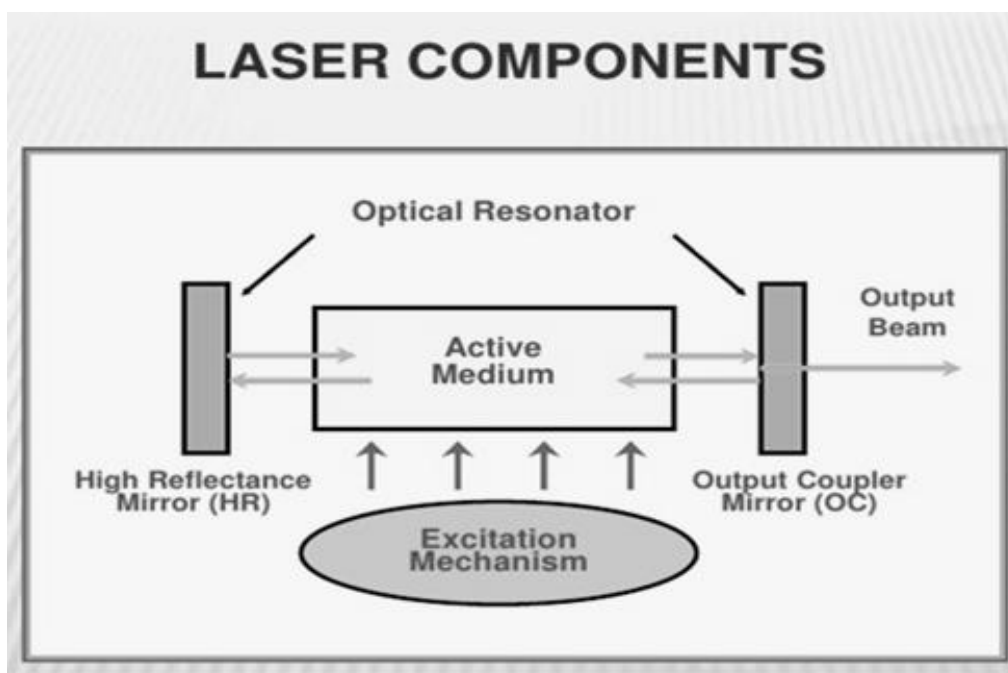
Ex: Ruby laser, Nd:YAG laser

➤ Electrical Pumping: By applying very high potential between the plates of discharge tube gas gets discharge leads to atom – atom collision and pumping.

Ex: He-Ne laser, CO₂ laser, Argon laser.

➤ Chemical Pumping: Exothermic chemical reactions liberate energy. This energy is used in pumping the atoms. Ex: Dye lasers.

[A suitable method of pumping is selected to produce population inversion. The choice of the method depends on the type of laser under consideration, the nature of the active medium and the laser output. For example, in solid state lasers, optical pumping method is used with a flash lamp as the source of light. Most often, the output from the source of light will be in the form of short flashes. The method may be suitable for pulsed lasers in which the output from the laser is also in pulses.]



2) An active medium: It is a medium in which the atoms or molecules will be excited to higher energy states in order to produce population inversion and lasing action occurs. The active medium that are used in lasers are solids, liquids, gases or semiconductors. Depending on the active medium used, lasers are classified into various types

- i) Gas lasers: Gases and mixture of gases. Ex: He-Ne laser, CO₂ laser.
- ii) Solid state lasers: Crystals and doped glasses. Ex: Ruby laser, Nd:YAG laser.
- iii) Liquids lasers: liquid medium, ex: SeOCl₂
- iv) Semiconductors lasers: semiconductor or diode. Ex: GaAs laser.

3) A laser cavity: An active medium bound between two mirrors form a laser cavity. One of the mirrors is fully reflecting and the other is partially transparent to allow some radiation to pass through. This cavity helps to increase the stimulated emission thereby increasing the intensity of the laser beam. There are two commonly used mirror configurations for optical cavity resonators. They are plane parallel mirrors and confocal mirrors. The radiation will propagate to and fro within the cavity between the two mirrors.

Conditions for Laser Action:

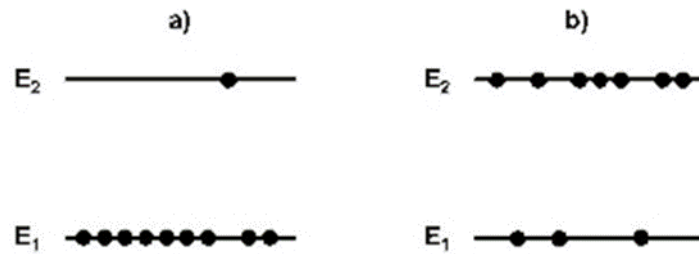
Efficient laser emission can be achieved under the following conditions:

- (i) Rate of stimulated emission should be more than the rate of spontaneous emission
- (ii) Rate of stimulated emission should be more than the rate of absorption.

Which can be achieved by

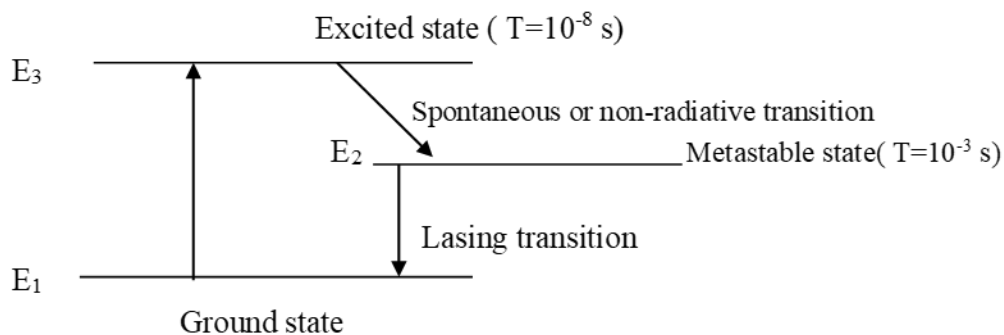
1) Population Inversion: Under normal condition the population of ground state is more than the excited state (Fig a). Hence the probability of absorption is more than the emission.

The population inversion is the state of a system at which the population (or number of electrons) of an excited state is higher than that in the lower state (Fig b). When population inversion is achieved there will be more number of emissions than absorptions. It is not possible to achieve population inversion with only two energy levels. The condition of population inversion can be achieved with the help of metastable states with which the atoms or molecules of the active medium are associated.



2) Metastable States: The lifetime of excited atom in ordinary excited states is of the order of 10^{-8} second which is very small. Within 10^{-8} second the atoms return to one of the lower energy states by spontaneous emission.

A metastable state is one which has a relatively longer lifetime which is of the order of 10^{-3} to 10^{-2} second and atoms excited to these levels will come down to lower levels at a much smaller rate than the rate at which they are excited. This property helps in achieving the population inversion.

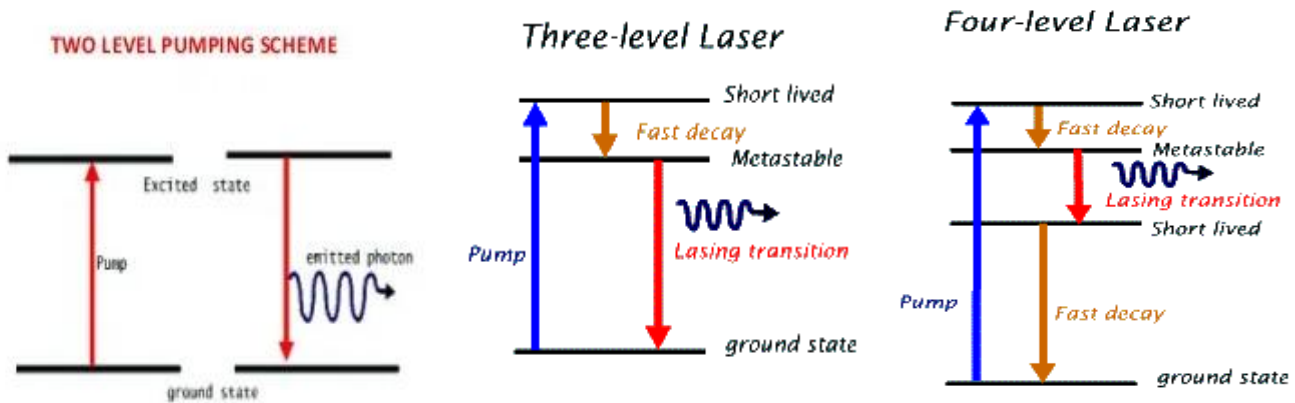


By supplying appropriate energy, the atoms are excited from E_1 to E_3 state. From E_3 state the atoms undergo spontaneous transition to E_1 and E_2 . But E_2 is a metastable state, where the atoms stay for longer duration. Hence population of which increases steadily. Under these conditions a stage will be reached wherein the population of E_2 exceeds that of E_1 , which is known as population inversion.

Two- Level, Three- Level and Four- Level Laser Systems:

The efficiency of laser emission depends on the nature of the active medium and the energy levels between which the laser action takes place.

For example, if there are only the two energy states, a ground state and a metastable state, it is said to be a **two-level system**.



Fig; Transitions in a two level, three level and four level lasers.

In the case of **two-level system**, the population inversion will not be possible. This is because, as more and more electrons are pumped to the metastable state, there will be more stimulated emission and at any instant of time, there will be no more than half the number of electrons in each state. As a result, no laser amplification occurs.

Population inversion will be possible only when the absorption is to a level higher than the metastable state as in the case of a **three-level laser**. Pumping takes the electrons from the ground state to the excited state. From this excited state, the electron makes a transition to the metastable state by spontaneous emission. The laser emission occurs when the electron in the metastable state is stimulated to undergo a transition to the ground state. However, a reverse transition is also possible from the ground state to the metastable state by a re absorption of the emitted radiation. This removes some photons from the laser beam and hence the efficiency of the laser decreases. This problem is overcome in a **four-level laser**.

In the case of a **four-level laser**, the laser transition occurs to an unstable intermediate state rather than the ground state. The electron decays to the ground state rapidly. Thus, the initial and the final laser levels are separated from the higher excited state and the ground state respectively. The probability of re absorption of laser radiation is thus avoided which improves the efficiency of the laser.

Nd:YAG LASER:

Construction: Neodymium-doped Yttrium Aluminium Garnet (Nd: YAG) laser is a solid-state laser in which Nd:YAG is used as a laser medium. It is a four-level laser system. These lasers operate in both pulsed and continuous mode. Nd stands for neodymium and YAG for Yttrium aluminium garnet ($\text{Y}_3\text{Al}_5\text{O}_{12}$). [It is developed by J.E. Geusic, H.M. Marcos and L.G. Van Viter in 1964]. The rod of $\text{Y}_3\text{Al}_5\text{O}_{12}$ is doped 1% with triply ionized (trivalent) neodymium. Nd^{3+} ions will replace the Y^{3+} ions in the crystal. The YAG crystal is transparent and colourless. When doped with approximately 1% Nd (1.38×10^{20} Nd ions per cm^3), the crystal takes a light blue colour. Maximum length of the rod is about 10 cm and diameter is 12 mm.

Active medium: The active medium or laser medium of the Nd:YAG laser is made up of a synthetic crystalline material Yttrium Aluminum Garnet (YAG) doped with Neodymium (Nd). Here Nd^{3+} ions act as active medium or active centers. YAG is just the host. That is the lasing transitions occur in the energy levels of Neodymium.

Pumping source: The pumping of Nd^{3+} ions to upper levels is done by krypton arc lamp. Xenon lamp can also be used as pumping source. Thus, the optical pumping is used to achieve population inversion.

Optical resonator system: Either the ends of the Nd: YAG rod are polished and silvered so as to act as the optical resonator system. Or the mirrors M1 (total reflector) & M2 (output coupler) can also act as a resonant cavity to produce stimulated emission and amplification process.

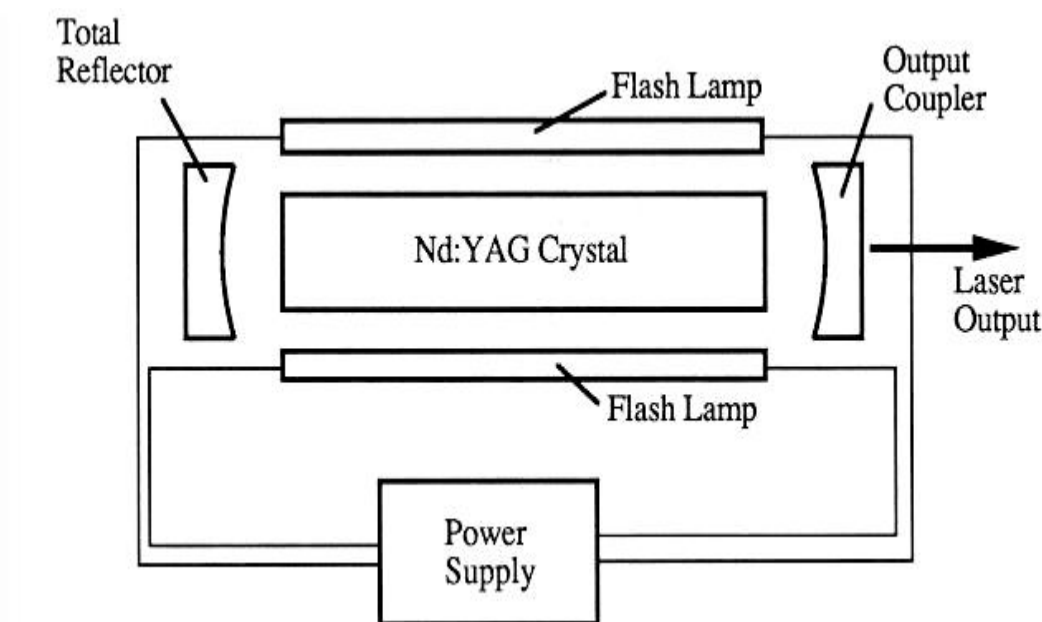


Figure: Schematic of Nd: YAG Laser

Working:

Nd:YAG is a four level laser system. The optical pumping is used in this laser. The pumping of neodymium (Nd^{3+}) ions to upper state (E4) is done by using krypton arc lamp. The wavelength of light of wavelength 720 nm to 800 nm excites the ground state (E1) Nd^{3+} ions to E4 states (lifetime $\sim$ ns). From E4 states, they make a non-radiative transition to E3 state. E3 is a metastable state (lifetime $\sim$ ms) so population inversion is achieved between the levels E3 and E2.

After this, the process of stimulated emission will occur. Thus, the laser emission will occur in between the levels E3 and E2 with the process of stimulated emission. So E3 is the upper laser level and E2 is the lower laser level. Then Nd^{3+} ions come back to the ground state E1. Laser emission will have wavelength of 1064 nm so occur in the infrared range of spectrum.

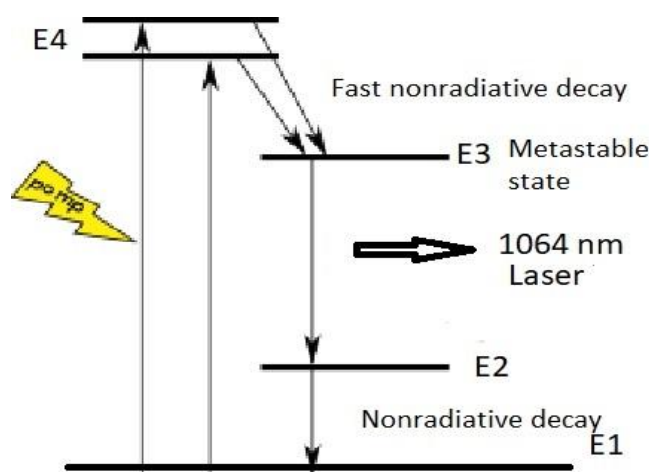


Figure: Energy level diagram of Nd:YAG laser

Wavelength and output of Nd: YAG laser: Nd: YAG laser has wavelength of about 1064 nm, that is in infrared region of spectrum. Nd YAG lasers may be operated in either a pulse mode or a continuous wave mode. The output powers for continuous wave mode range from 0.1 watts to 100 watts. The peak power for the pulse mode can range from tens or hundreds of kilowatts with a pulse lasting milliseconds. The pulse will last for 10-20 nanoseconds for a peak power of hundred megawatts.

Applications of Nd: YAG laser:

Nd:YAG laser has following applications or uses or advantages:

Advantages

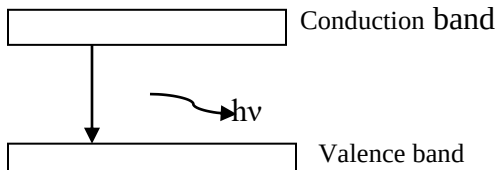
- They produce continuous laser at room temperature
- They can be used as portable systems since the rods are small.

Applications

- They have surgical applications.
- They are used material processing such as drilling, spot welding and marking.
- They are used as pumping tunable visible light lasers.
- They have applications in military such as including range finders and target designators.
- Research applications such as Raman spectroscopy, remote sensing, mass spectrometry.

Semiconductor Laser:

Semiconductor laser is the one in which the active medium is formulated by semiconducting materials. They are smallest in size and least expensive. They are used as photonic sources in optical communication. Only a kind of semiconductors called direct band gap semiconductors are employed in semiconductor lasers.



When junction is forward biased at low current electron hole pair recombines to give out a photon. This is spontaneous emission. When current crosses a threshold value a large number of electron hole pair occurs at the junction. This is population inversion. The recombination of these electron hole produces lasing action. The photons emitted will come out of the junction edge. Gallium Arsenide is a class of direct band gap semiconductor.

Gallium Arsenide (GaAs) Laser:

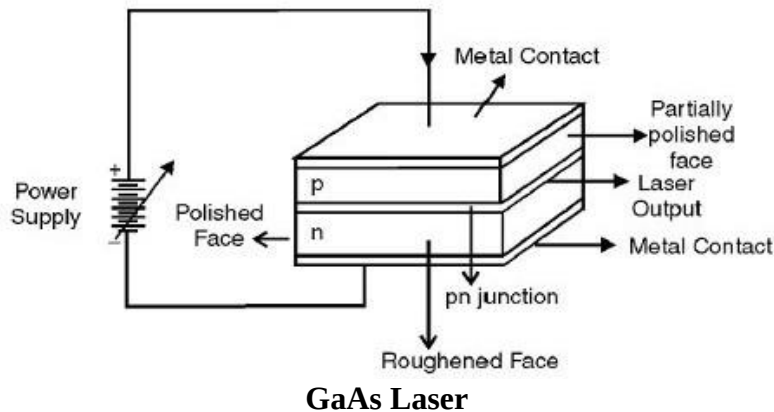
Principle: GaAs is a direct band gap semiconductor. In a **direct band gap semiconductor** during the recombination of electron and hole, the recombination energy is given out as photon.

In a diode laser, the p-n junction is formed using degenerate semiconductors. When the dopant concentration is increased, the impurity levels will expand into bands and then merge with the nearest energy band. In an n-type semiconductor, the donor band overlaps with the conduction band and for a p-type semiconductor the acceptor band overlaps with the valence band. Such semiconductors are called **degenerate semiconductors**.

When p-n junctions made out of these degenerate semiconductors are forward biased, electrons and holes are injected across the junction in large concentration. This results in a condition identical to population inversion in the region close to the junction and transitions from the bottom of the conduction band to the top of the valence band can take place with the emission of radiation of energy equal to the band gap energy.

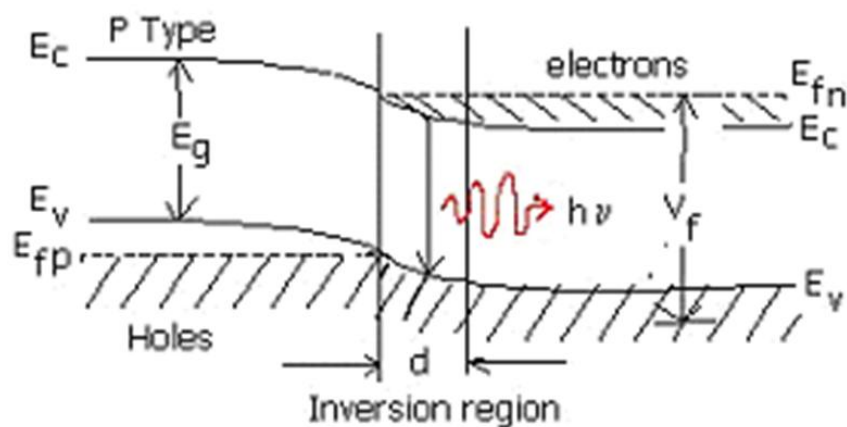
$$E_g = hv = hc/\lambda \quad \text{or} \quad \lambda = hc/E_g$$

Construction: GaAs laser diode is a single crystal of GaAs and consists of heavily doped n and p sections. The n- section is formed by doping with Tellurium and p- section is formed by doping with Zinc. [The doping concentration is very high is of the order of 10^{17} to 10^{19} dopant atoms/cm³]. A pair of parallel planes of the crystal are cleaved or polished at right angle to the p-n layer. These planes play the role of reflecting mirror. The other two remaining sides perpendicular to the junction are roughened to suppress reflection of the photons. The end surfaces of the p and n sections parallel to the plane of the junction are provided with electrodes for the application of forward bias voltage.



Working: GaAs laser diode is subjected to a forward bias as shown in figure. Due to the forward bias, the holes are injected into the p-side of the junction and electrons are injected into the n-side. During the flow when a hole meets an electron recombination takes place resulting in the emission of a photon. In a GaAs laser by heavy doping, a large number of electrons are available in an n- type material and a large number of holes are available in a p-type material.

When the current in the diode is low, the concentration of the electrons at the bottom of conduction band will be still lesser than that in the valence band, and the recombination result in only spontaneous emissions. But as the current is increased, a threshold for lasing will be attained beyond which a population inversion can be achieved between the conduction band and valence band and an active region is formulated very near the junction. At this stage a photon released by a spontaneous emission may trigger stimulated emission over a large number of recombination, leading to the buildup of laser radiation of high power.



Energy band diagram of a diode laser under forward bias

Since the energy gap of GaAs is 1.4 eV, the wavelength of the emitted light is

$$\lambda = hc/E_g = 8400 \text{ \AA}$$

Applications:

Semiconductor lasers are the smallest and least expensive of all the lasers available.

- They find important use in optical communications
- They are also used extensively as reading devices for compact disc (CD) players, bar code readers.
- They are used in Laser printers, Laser Pointers etc.

APPLICATIONS OF LASERS:

- One of the major applications is in pure science to investigate the interaction of matter with intense electromagnetic radiation. The directionality and coherence of the laser beam are useful in the measurement of distances based on interferometric methods.
- In the field of communication, laser has been used with proper modulation for information transmission.
- Because of the high intensity and energy associated with laser beams, they can be utilized for applications such as welding, cutting and ablation of materials.
- Laser has also been used for medical applications like treatment of dental decay, destruction of tumors, treatment of skin diseases and eye surgery.
- Holography is another important application which helps in recording the amplitude as well as the phase of light reflected from objects thereby preserving the three-dimensional information.

Laser	Wavelength (nm)	Power	Sample applications
Excimer	190–350 (UV)	<1–20 watts	Ophthalmological surgery (PRK and LASIK)
Dye	400–800	<1–5 watts	Removal of port-wine stains and pigmented lesions
Nd:YAG	1064 (IR), with frequency doubling: 532 (green)/355 (UV)	<1 to >100 watts	General surgery, ophthalmological surgery; lithotripsy
Argon (CW)	514 (green) and 488 (blue-green)	<1 to 20 watts	Surgery for glaucoma
Copper vapor	578 (yellow) and 511 (green)	≤1 watt	Ophthalmological surgery
Gold vapor	630 (red)	≤1 watt	Photodynamic therapy, Ophthalmological surgery
Helium neon (CW)	632.8 (red)	<1–25 milliwatts	Ophthalmological surgery
Krypton (CW)	676.4 (red)	≤5 watts	Ophthalmological surgery
Ruby (pulsed)	694 (red)	≤10 watts	Tattoo removal
Diode lasers	750–1550 (red and near IR)	≤10 watts	Photocoagulation for general surgery, hair removal
Erbium:YAG (pulsed)	1540 (IR)		Hair removal, skin resurfacing
Holmium:YAG (pulsed)	2140 (IR)	≤60 watts	General surgery
Carbon dioxide (CW or pulsed)	10,600 (IR)	<100 watts	General surgery, skin resurfacing

Application of Laser in Barcode scanner and Laser Printer:

1. Laser Barcode Scanner:

7.17 LASER BAR CODE SCANNER

The Universal Product Code (UPC) and European Article Numbering (EAN) systems are widely used for identifying any product, its manufacturer, its price and other specifications. A special form of the EAN code is the International Standard Book Numbering (ISBN) system for identifying books. A bar code consists of a series of strips of dark and white bands (Fig. 7.93). These white and dark bands are of different widths separated from each other by specific distances that contain all the information about the product. A laser is used to scan the bar code with the help of a rotating mirror (Fig. 7.94). Typical scanning speeds are about 200 m/s. Such high speeds enable the product to be scanned even when in motion in a delivery line. When the laser beam is incident on the bar code, the amount of light scattered depends on whether the strip is black or white. Since the bars are separated by variable distance, light intensity varies with time and is recorded by the photodetector. The signal is fed into an amplifier and later to a decoder which displays the information on a TV monitor and also sends it to product inventory system.

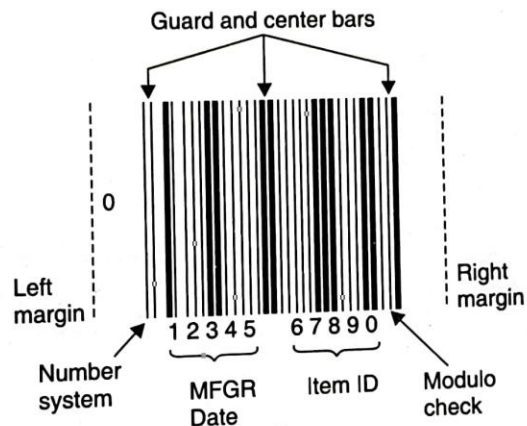


Fig. 7.93 Bar code

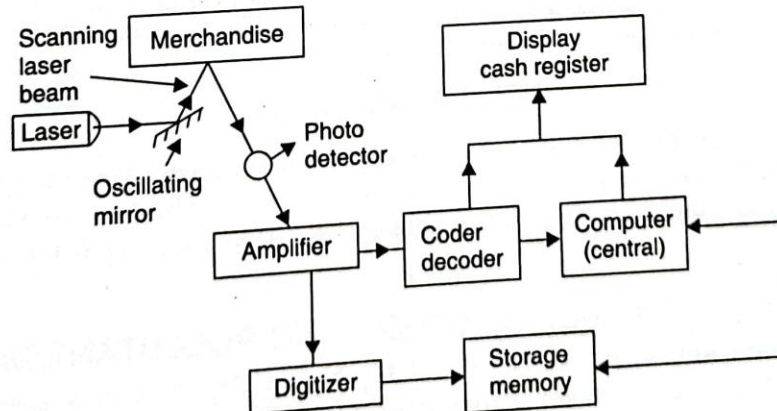


Fig. 7.94 Laser bar code scanner

2. Laser Printer:

7.18 LASER PRINTER

A laser printer is essentially a xerographic device (Fig. 7.95). The photoconductor in a xerographic copier allows light to control the distribution of electric charge. An optical image of the original document is projected on to the charged region of the photoconductor. Wherever light falls, the photoconductor conducts electricity and the charge on that region escapes. The result is a charge image on the surface of the photoconductor. Tiny black toner particles, oppositely charged from the unilluminated portions of the photoconductor are brought near the charge image. These toner particles stick to the charged portions of the photoconductor, forming a visible image of the document. In a laser printer, a laser beam is used to write a charge image directly onto the photoconductor drum. This rotating drum is charged by a coratron and then a spinning mirror scans laser light rapidly across the surface. Wherever laser light falls on the drum charge flows through the photoconductor. A computer in the printer turns the laser on and off as it systematically constructs the charge image one dot at a time. The printer then sticks the toner particles to the charge image and transfers it to the paper. Most laser printers use special electrostatic tricks to reverse the charge image, so that the toner particles stick only to the portions of the drum that were exposed to the laser light. This reversal makes it easier for the printer to produce white backgrounds for its prints.

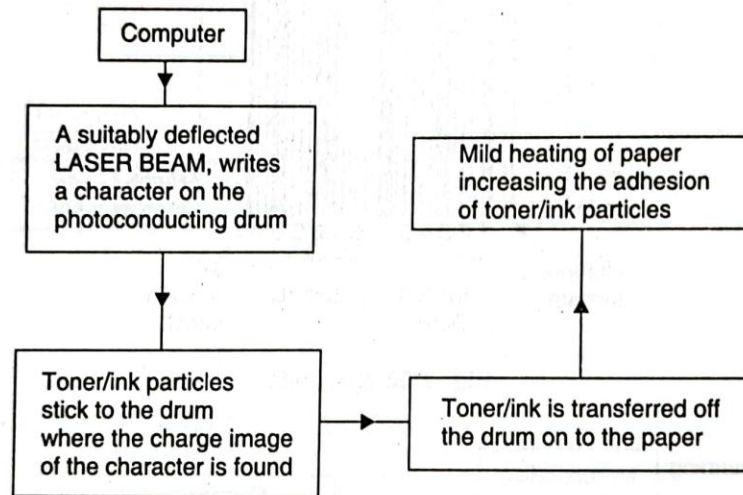
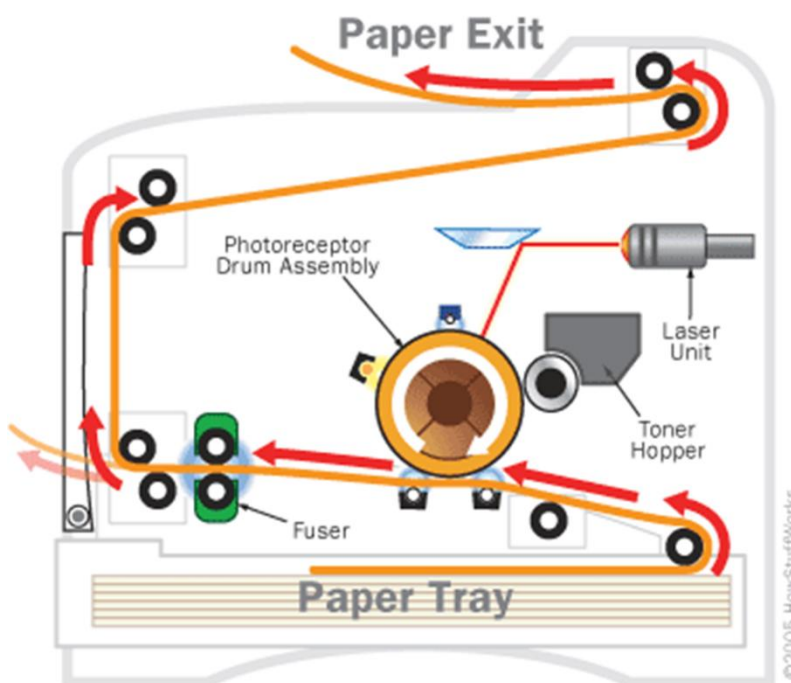


Fig. 7.95 Laser printer

A colour laser printer works like a single colour model, except that the process is repeated four times and a different toner colour is used for each pass. The four colours used are cyan, magenta, yellow and black. Laser printers have resolution ranging from 300 dots per inch (dpi) to 1800 dpi.



NUMERICAL EXAMPLES

1. A He-Ne laser emits light at a wavelength of 632.8 nm and has an output power of 5 mW. How many photons are emitted each second by this laser?

Solution:

Wavelength of emitted light $\lambda = 632.8 \times 10^{-9} \text{ m}$.

Energy of each photon $\Delta E = hc/\lambda = (6.62 \times 10^{-34} \times 3 \times 10^8) / 632.8 \times 10^{-9}$

$$= 3.138 \times 10^{-19} \text{ J}$$

Energy emitted by the laser, $E = 5 \text{ mW} = 5 \times 10^{-3} \text{ Js}^{-1}$

Number of photons emitted $= E/\Delta E = (5 \times 10^{-3}) / (3.138 \times 10^{-19}) = 1.59 \times 10^{16} \text{ photons/sec}$

2. Find the ratio of population of the two energy states the transition between which is responsible for the emission of photons of wavelength 694.3 nm. Assume the ambient temperature as 27 °C.

Solution:

Energy of the emitted radiation $= (E_2 - E_1) = hc/\lambda = (6.63 \times 10^{-34} \times 3 \times 10^8) / 694.3 \times 10^{-9}$
 $= 2.86 \times 10^{-19} \text{ J}$

The ratio of population (N_2/N_1) between two energy states E_2 and E_1 is given by the Boltzman relation,

$$(N_2/N_1) = \exp[-(E_2 - E_1)/kT] = \exp - [(2.86 \times 10^{-19}) / (1.38 \times 10^{-23} \times 300)]$$

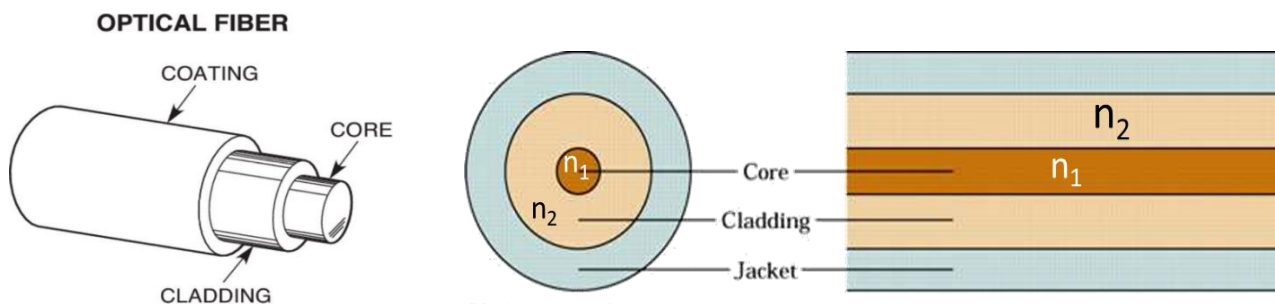
$$= \exp(-69.08)$$

$$= 9.98 \times 10^{-31}$$

3. A He- Ne laser emits light at a wavelength of 632.8 nm and has an output power of 2.3 mW. How many photons are emitted in each minute by this laser when operated?
4. The ratio of population of two energy levels in a laser system at 320 K is 10^{-30} . Find the wavelength of the radiation emitted.
5. A laser operating at 632.8 nm emits 3.18×10^{19} photons per second. Calculate the output power of the laser if the input power is 100 watts. Also find the percentage power converted into coherent light energy.
6. A Nd:YAG laser emits pulse of power of 1W and of duration 12ps . Calculate the number of photons in each pulse, if the laser wavelength is 1064nm.
7. The ratio of population of two energy levels is 10^{-30} . Determine the temperature if the lasing wavelength is 48.6 μm .

OPTICAL FIBERS

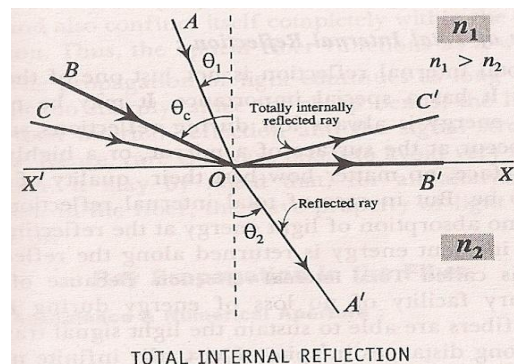
Optical Fibers: An optical fiber is essentially a waveguide for light. It is long, thin, cylindrical in shape and it consists of mainly two regions. The central cylindrical region is the light guiding region known as the **Core**. The coaxial region surrounding the core is known as **Cladding**. The **refractive index of the core material is higher than that of the cladding ($n_1 > n_2$)**. The core and cladding are made of either **glass or plastic**. The outermost region surrounding cladding is called the **sheath or protective buffer coating**. Additional covering is provided to improve the strength and also to protect it from moisture and mechanical wear. With all the protective coverings, the optical fibers are still very thin, less than 0.1 mm in diameter typically.



[There are mainly two types of materials used for the construction of optical fibers: glass and plastic. The glass fibers are made of silicates like sodium borosilicate, sodium calcium silicate and lead silicate. Pure silica (SiO_2) has a refractive index of 1.45 at a wavelength of $1\text{ }\mu\text{m}$. Additives like B_2O_3 can be used to lower the R.I. while additives like GeO_2 can be used to raise it. Hence, a typical fiber may have $\text{SiO}_2\text{:GeO}_2$ as the core and pure SiO_2 as cladding. These glass fibers may be manufactured with a wide range of refractive indices and are known for their characteristics of low scattering. They are good for communication applications. Fibers can also be made from plastic, but these fibers are associated with high attenuation due to scattering. Alternatively, plastic coated silica fibers may be made with core of pure silica and cladded with a suitable polymer. These optical fibers are economical but are suitable for medium distances with moderate bandwidth.]

Total Internal Reflection (TIR) :

Propagation of light through an optical fiber is by total internal reflection (TIR). For a light ray traveling from one medium of refractive index n_1 to a second medium of refractive index n_2 , we have, by Snell's law, $n_1 \sin \theta_1 = n_2 \sin \theta_2$, where θ_1 and θ_2 are the angle of incidence and the angle of refraction respectively.



When the ray of light is traveling from a denser medium to a rarer medium, i.e., $n_1 > n_2$, the ray bends away from the normal as shown in figure (ray AOA'). If the angle of incidence is increased, for a particular angle of incidence called critical angle θ_c , the refracted ray just grazes the interface, (ray BOB') i.e., $\theta_2 = 90^\circ$.

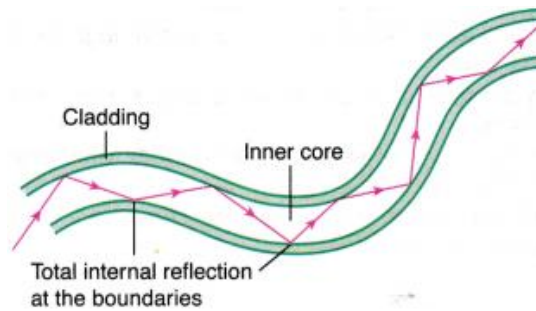
Hence the critical angle of incidence θ_c is given by

$$n_1 \sin \theta_c = n_2 \sin 90^\circ$$

or **$\sin \theta_c = n_2/n_1$**

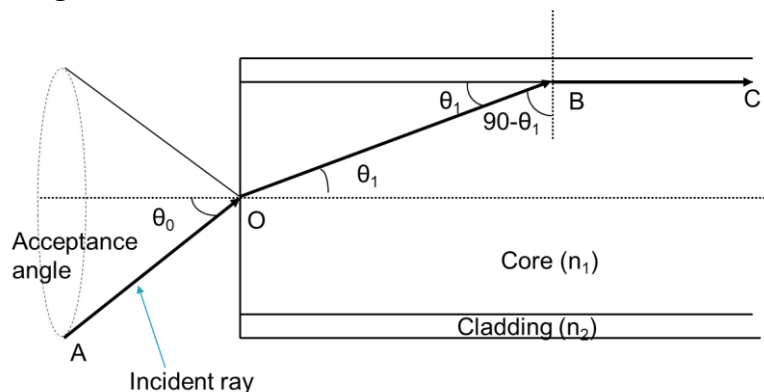
The ray will be totally internally reflected into the denser medium when the angle of incidence is greater than the critical angle (known as total internal reflection) (ray COC').

Thus, if we have a fiber consisting of a core of higher R.I. surrounded by a cladding of lower R.I., the light ray once enters into the core, it will propagate by means of multiple total internal reflection at core-cladding interface.



Ray propagation in the optical fiber: Angle of Acceptance, Acceptance cone & Numerical Aperture

Consider a ray of light AO enters into the core of an optical fiber at an angle θ_0 to the fiber axis as shown in figure. This ray is refracted along OB at an angle θ_1 in the core and further proceeds to fall at critical angle $(90-\theta_1)$ at B on the core-cladding interface. Hence the ray is refracted at 90° to the normal to the interface, i.e. it grazes along BC.



It is clear from the figure that, any ray that enters the core at an angle less than θ_0 , will have angle of incidence $(90-\theta_1)$ at the interface greater than the critical angle and hence undergoes total internal reflection.

On the other hand, any ray that enters at an angle greater than θ_0 at O, will have angle of incidence less than the critical angle at the interface. Hence it refracts into the cladding and is lost from the core.

The angle θ_0 is called **angle of acceptance**. **Angle of acceptance** (semi-angle of the cone of acceptance) is the maximum allowed angle that the incident ray can make with the fiber axis for transmission through the fiber by total internal reflection.

If the incident ray AO is rotated around the fiber axis keeping θ_0 same, then it describes a conical surface called **acceptance cone**. Only those rays which are funneled into the fiber within this cone will propagate through the fiber by TIR.

Numerical Aperture (N.A.) is a measure of light-gathering or light-collecting ability of an optical fiber. The numerical aperture of an optical fiber is given by,

$$NA = n_0 \sin \theta_0$$

Where n_0 is the refractive index of the surrounding medium. Usually the surrounding medium is air, and for air, n_0 is 1.0003 which is close to 1, the expression can be approximated to

$$NA = \sin \theta_0$$

Expression for Numerical Aperture:

Let n_0 , n_1 and n_2 be the refractive indices of surrounding medium, core and cladding respectively.

By applying Snell's law for refraction at the point of entrance at O,

$$n_0 \sin \theta_0 = n_1 \sin \theta_1$$

$$n_0 \sin \theta_0 = n_1 \sqrt{1 - \cos^2 \theta_1} \quad \text{-----(1)}$$

Applying Snell's law for refraction at B on the interface,

$$n_1 \sin (90 - \theta_1) = n_2 \sin 90^\circ$$

$$\text{or} \quad n_1 \cos \theta_1 = n_2$$

$$\text{or} \quad \cos \theta_1 = n_2 / n_1 \quad \text{-----(2)}$$

Substituting (2) in (1)

$$n_0 \sin \theta_0 = n_1 \sqrt{1 - \cos^2 \theta_1} = n_1 \sqrt{1 - \frac{n_2^2}{n_1^2}} = \sqrt{(n_1^2 - n_2^2)}$$

$$\therefore N.A. = n_0 \sin \theta_0 = \sqrt{(n_1^2 - n_2^2)}$$

If the surrounding medium is air, then, $n_0 = 1$.

Then

$$N.A. = \sin \theta_0 = \sqrt{n_1^2 - n_2^2}$$

If θ_i is the angle of incidence of an incident ray, then the condition for ray propagation is

$$\theta_i < \theta_0$$

$$\text{or} \quad \sin \theta_i < \sin \theta_0$$

$$\text{or} \quad \sin \theta_i < N.A.$$

Fractional index Change or Relative Refractive index difference (Δ):

It is the ratio of the refractive index difference between core and cladding to the refractive index of the core of an optical fiber.

i.e. $\Delta = \frac{n_1 - n_2}{n_1}$

Relation between N.A. and Δ :

$$\Delta = \frac{n_1 - n_2}{n_1} \quad \text{or} \quad n_1 - n_2 = n_1 \Delta$$

$$N.A. = \sqrt{n_1^2 - n_2^2} = \sqrt{(n_1 + n_2)(n_1 - n_2)} = \sqrt{(n_1 + n_2)n_1 \Delta}$$

Since $n_1 \approx n_2$, $(n_1 + n_2) = 2n_1$

Hence $N.A. = \sqrt{2n_1^2 \Delta} = n_1 \sqrt{2\Delta}$

Modes of propagation in a fiber:

There are various modes by which light waves can be made to propagate in an optical fiber. The mode of propagation depends on the construction of the optical fiber. Modes of propagation refers to the number of paths for the light rays along which the waves are in phase inside the fiber. Multimode means several paths are available. Single mode means single ray propagation.

V-Number:

It is a parameter which determines the number of modes of propagation.

$$V = \frac{\pi d}{\lambda} NA$$

$$V = \frac{2\pi r}{\lambda} \sqrt{n_1^2 - n_2^2} = \frac{2\pi r}{\lambda} NA$$

Where d-diameter of the core and r-radius of the core. λ is wavelength of light.

Number of modes $M \approx \frac{V^2}{2} \approx \frac{\pi^2 d^2}{2\lambda^2} (n_1^2 - n_2^2)$

Types of Optical Fibers:

Classification of optical fibers based on material used:

- i. Glass fibers
- ii. Plastic fibers

Classification of optical fibers based on modes of propagation:

- i. Single mode fibers
- ii. Multi- mode fibers

Optical fibers are classified into two types based on the refractive index profile:

- i. Step index fibers
- ii. Graded index fibers.

Optical fibers are broadly classified into three types:

- (1) Step index Single mode fibers,
- (2) Step index Multi mode fibers, and
- (3) Graded index Multi mode fibers.

Refractive Index Profile:

The refractive index profile of an optical fiber is a graphical representation of the magnitude of the refractive index across the fiber.

The refractive index is plotted on the vertical axis, and the radial distance from the core axis is plotted on the horizontal axis. The boundary between the core and cladding may either be abrupt, in step-index fiber, or gradual, in graded-index fiber.

- In **step-index** fibers the refractive index changes abruptly at the interface between core and cladding
- **Graded-index** fiber, the refractive index decreases gradually away from the center of the core.



(1)Single mode fibers (SMF)

A step-index fiber has a central core with a uniform refractive index and outside cladding surrounding the core also has a uniform refractive index; however, the refractive index of the cladding is less than that of the central core. (Fig. 1). The R.I. profile is a step-index profile as the refractive index changes abruptly at the interface between the core and the cladding.

The diameter of the core is very small about 8-10 μm and external diameter of cladding is 60-70 μm . In a single mode fiber, the core is so small that there is a single path for the light to propagate in the fiber. Only laser can be used as a source of light.

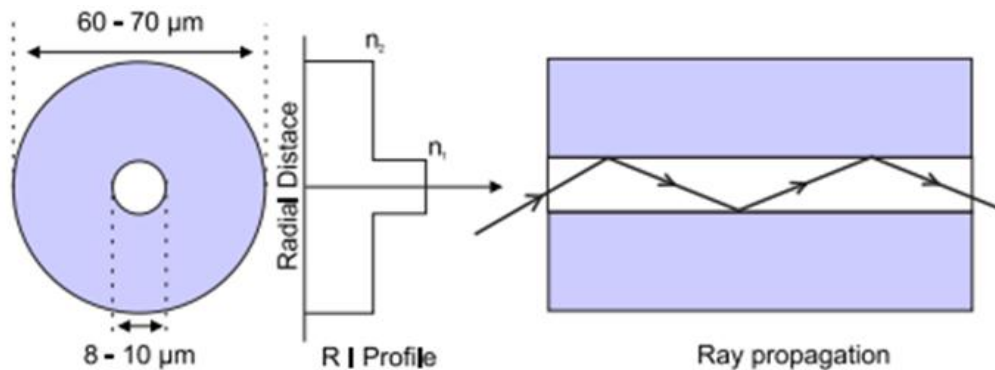


Fig. 1: Single mode step index fiber

Applications:

- Nearly 80% of the fibers manufactured are of this type.
- It is used for long distance communication
- It finds particular application in submarine cable system..

Advantages:

- It can transmit signal over longer distances.
- The signal propagating through fiber is free from intermodal dispersion.

Disadvantages:

- It is difficult to launch the light through the fiber
- The single mode optical fibers are costly, because the fabrication is difficult.
- It is difficult to achieve end to end connection of fibers.

(2) Step Index Multi mode fibers:

A step-index Multi mode fibers has a central core with a uniform refractive index and outside cladding also has a uniform refractive index; however, the refractive index of the cladding is less than that of the central core. (Fig. 1). The refractive index changes abruptly at the interface between the core and the cladding and hence the R.I. profile is a step-index profile.

The diameter of the core is $50 - 200 \mu\text{m}$ and external diameter of cladding is $100 - 250 \mu\text{m}$. The fiber supports many modes for propagation. Either a LASER or LED can be used as source of light.

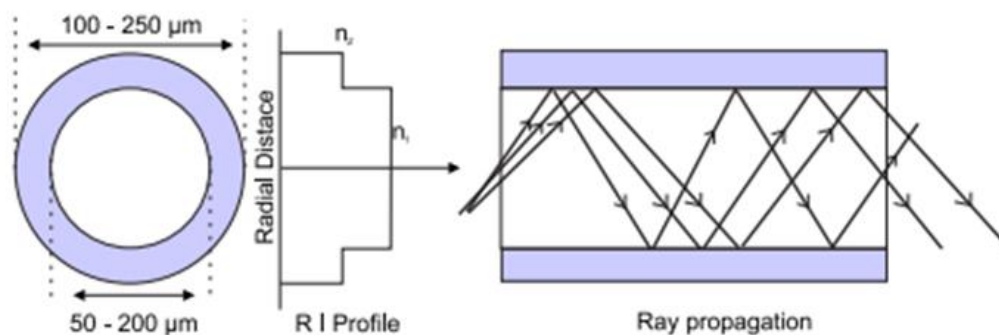


Fig. 2: Step index multimode fiber

Application - They are used in data links.

Advantages:

- Can transmit more than one mode
- Launching light into the fiber is easy
- The multi-mode optical fibers are cheaper than single mode fibers, because the fabrication is easy.

Disadvantages:

- It is used to transmit signal only for short distances.
- Pulse distortion is more.
- Multimode fibers suffer intermodal dispersion.

(iii) Graded index Multi mode fibers:

In a graded index multimode fiber, the refractive index of the core is maximum at the center of the core and decreases gradually and becomes equal to that of the cladding at the interface between core and cladding (Fig.1). As the refractive index decreases gradually away from the center of the core, the R. I. profile is a graded index profile.

Diameter of the core is 50-200 μm , Cladding diameter is 100-250 μm . Hence The fiber supports many modes for propagation. Either a LASER or LED can be used as source of light.

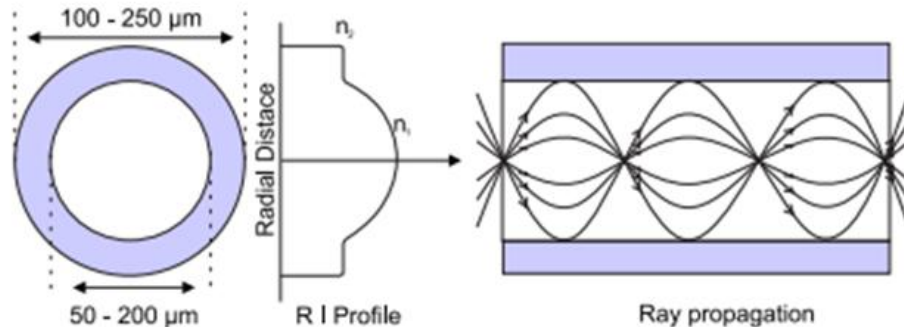


Fig. 3: Graded index multimode fiber

Application – Generally they are used in comparatively less bandwidth and short-distance applications like LANs (Local Area Networks) and in telephone trunks between central offices.

Advantages:

- The information carrying capacity is more.
- Pulse distortion is less.
- Intermodal dispersion less due to variation in RI.

Disadvantage: It is expensive compared to step-index fiber due to difficulty in manufacturing. It has less light coupling efficiency.

Signal distortion in optical fibers:

Propagation of a signal through the optical fiber involves total internal reflection of light rays many times. Further, the rays are reflected at various angles. The rays reflected at higher angles travel greater distances than the rays reflected at low angles. As a result, all the rays do not arrive at the end of the fiber simultaneously and the light pulse broadens as it travels through the fiber. Since the output pulse does not match with the input pulse; the **signal is said to be distorted**. When white light source is used instead of a monochromatic radiation, another kind of distortion occurs. Since radiation of different wavelengths has different velocities, they do not arrive at the output simultaneously. This distortion is called **chromatic dispersion**.

The signal distortion is quite considerable in multimode step index fibers. In graded index fibers, the light travels with different velocities in different parts of the core as the refractive index varies radially along the core. The rays travel faster near the outer edge of the core (since the refractive index is smaller) than near the center. Hence, all the rays arrive at the output almost at the same time and the signal distortion is reduced. In a single mode step index fiber also the distortion is less than that in multi-mode step index fibers.

Signal attenuation in optical fibers:

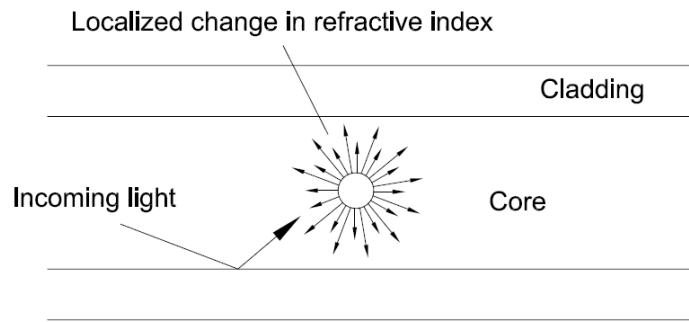
Attenuation is the loss of optical power suffered by the optical signal as it propagates through the fiber. The main reason for the loss of light intensity over the length of the fiber is due to (1) Light Absorption, (2) Scattering and (3) Radiation Losses.

(1) Light Absorption (Absorption losses): Absorption loss is due to the loss of signal power by absorption of photons associated with the signal. It takes place by two ways.

(i) Absorption by impurities or extrinsic: Intrinsic attenuation is caused by impurities incorporated into the glass during the manufacturing process. The presence of impurities like copper, chromium, iron etc (ie, transition metals) and hydroxy ions (OH) will absorb some photons in the signal. The photons may be absorbed by the impurity atoms and re-emitted by spontaneous emission or converted into some other form of energy.

(ii) Intrinsic Absorption: Even if the material has no impurities, the fiber material itself may absorb some photons. This is called intrinsic absorption. Intrinsic attenuation can also occur due to absorption of signal in the fiber.

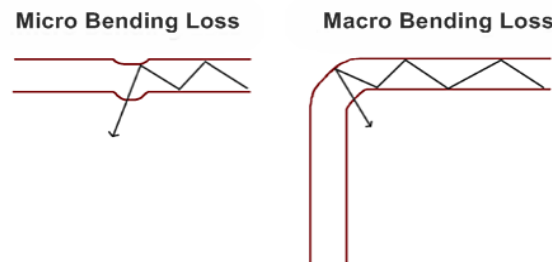
(2) Scattering losses: This loss is due to the scattering of signal photons by some compositional inhomogeneity's in the fiber. Since glass is a heterogeneous mixture of many oxides like SiO_2 , P_2O_5 (Phosphorous pentoxide) etc., the composition of the molecular distribution varies from point to point. Hence due to the inhomogeneity in the material there will be sharp variation in refractive index value inside the glass over a distance that is smaller than wavelength of light. The scattering of photons by such scattering centers is same as Rayleigh scattering and it is inversely proportional to the 4th power of wavelength. The presence of structural inhomogeneity's (defects, trapped gas bubbles, unreacted starting materials etc.) can also scatter some photons.



(3) Radiation Losses (Bending losses): This is caused by two mechanisms, namely, macroscopic bending and microscopic bending of the fiber.

Macroscopic bending: Because of the bending of the fiber during installation, which are known as *macroscopic bending*, the curvature in the fiber also leads to attenuation. All optical fibers have a minimum bend radius specification that should not be exceeded. This is a restriction on how much bend a fiber can withstand before experiencing problems in optical performance or mechanical reliability. A bend in the fiber may result in the modification of the angle of incidence on the core-cladding interface and hence may lead to signal loss.

Microscopic bending is a small-scale distortion of the core-cladding interface in a localized region. This may be related to pressure, temperature, tensile stress or crushing force to which the fiber might have been subjected to.



Attenuation coefficient (α):

Attenuation is the loss of optical power as light travels through a fiber. It is measured in decibels (dB/km). Over a set distance, a fiber with a lower attenuation will allow more power to reach its receiver than a fiber with higher attenuation. The attenuation coefficient (constant) α of the fiber, in units of dB/km, is given by

$$\alpha = -\frac{10}{L} \log_{10} \left(\frac{P_{out}}{P_{in}} \right) \quad \text{dB/km}$$

Where L - Length of the fiber in km

P_{in} – Input power in watt

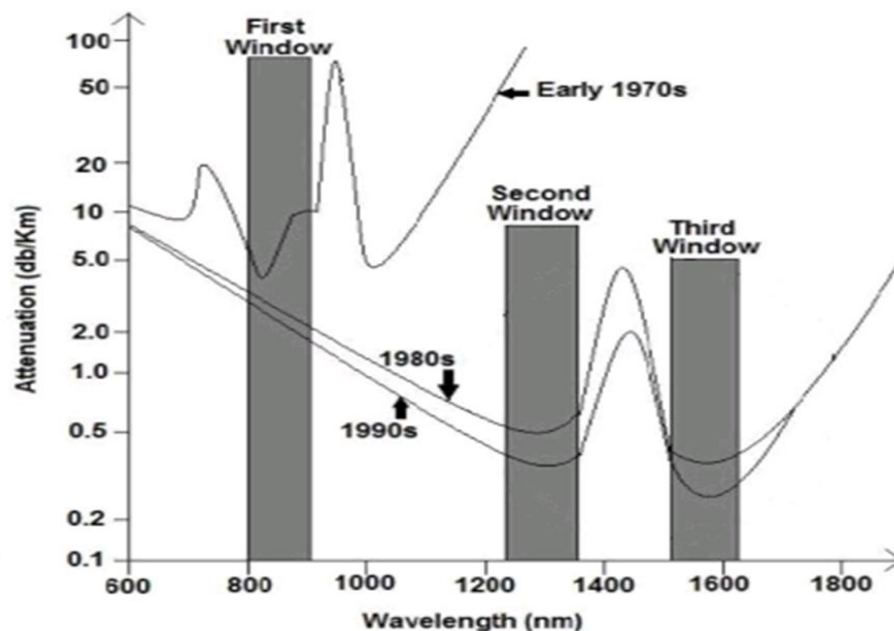
P_{out} – Output power in watt

Applications of optical fibers:

Optical fibers find extensive applications in

- Fiber optic communication – Telecommunications, Local Area Networks, Cable TV, CCTV
- Optical Fiber Sensor
- Applications in medicine - endoscopy
- Application in industry

Attenuation Spectrum of an Optical Fiber – Optical Window:

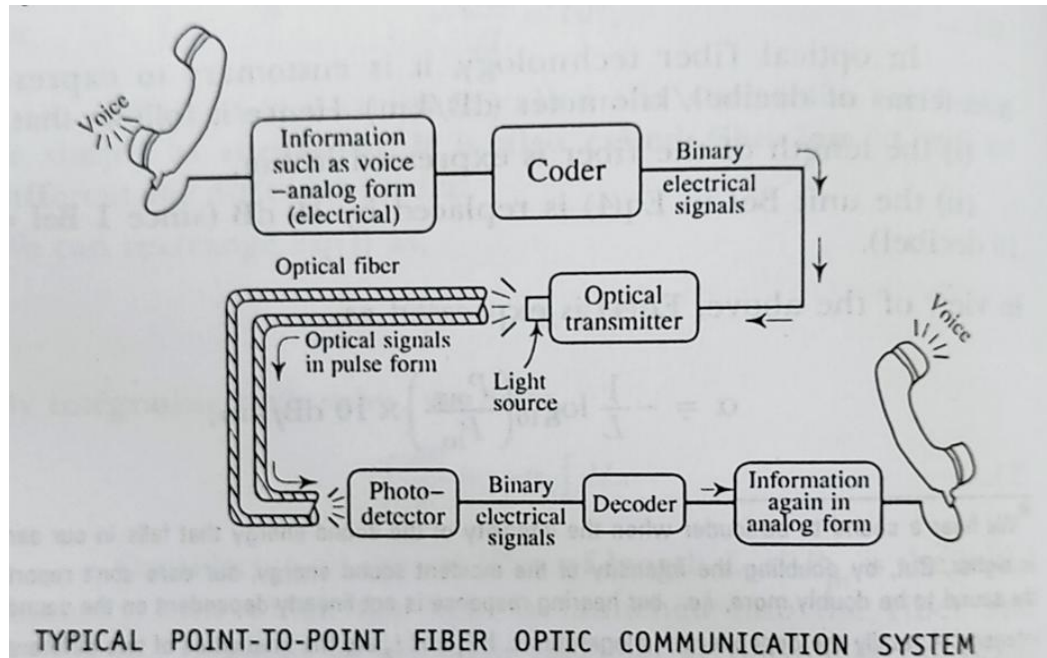


- In case of optical transmission, the loss is wavelength dependent. So, there is a specific band of wavelength where the signal attenuation is minimum which is known as optical or operating window. The wavelength of operation from the optical window is selected as they offer minimum attenuation.
- Figure above shows three optical windows which offer minimum signal attenuation and also relationship between attenuation and wavelength.
- The first optical window is defined from 800-900 nm, where the minimum signal loss is 4dB/km. In early 1970's this window was used for operation of optical sources and detectors.
- By reducing the concentration of hydroxyl ions and metallic impurities in the fiber material, in 1980's manufacturers were able to fabricate optical fibers with very low loss in the 1100-1600 nm region. This spectral band is called long wavelength region.
- The second optical window is centered at 1310 nm also called O-band, which offers 0.5dB/km.
- The third optical window is centered at 1550 nm also called C-band, which gives the loss of 0.2dB/km.
- Hence while designing optical systems for long distance applications the 1550 nm wavelength is preferred because loss offered at this wavelength is minimum than any other wavelength.

Point to point communication system:

Optical fiber communication is the transmission of information by propagation of optical signal through optical fibers over the required distance, which involves deriving optical signal from electrical signal at the transmitting end, and conversion of optical signal back to electrical signal at the receiving end.

As a simple example we can consider the basics of point to point communication system.



In a typical point to point communication system, we have analog information such as voice of a telephone user. The voice gives rise to electrical signals in analog form coming out of the transmitter section of the telephone. The analog signal is converted to binary data with the help of an electronic system called coder. The binary data comes out as a stream of electrical pulses from the coder. These electrical pulses are converted into pulses of optical power by modulating the light emitted by an optical source (such as an LED or laser diode) in the binary form. This unit is called an optical transmitter (Fig. 11) from which the optical power is fed into the fiber.

Out of the incident light which is funnelled into the core within the half angle acceptance cone, only certain modes will be sustained for propagation within the fiber by means of total internal reflection. As it propagates, the signal is subjected to both attenuation and delay distortion. Delay distortion is the reduction in the quality of signal because of spreading of pulses with time. The pulse spreading is mainly due to the variation in velocity of various spectral components of the pulse during its propagation in the fiber. Because of such an effect, the pulses which are initially separated, start overlapping in time.

These effects cause continuous degradation of the signal as the light propagates, and may reach a limiting stage beyond which it may not be possible to retrieve the information from the light signal. Just at this limiting stage, a repeater is needed in the transmission path. An optical repeater consists of a receiver and a transmitter arranged adjacently. The receiver section converts the optical signal into corresponding electrical signal. Further, this electrical signal is amplified, and recast in the original form by means of an electrical regenerator, which is also a part of the receiver section. This reshaped electrical signal which is in the form of binary data stream, is sent into an optical transmitter section where the electrical signal is again converted back to optical signal, and then fed back into the optical fiber line.

Finally, at the receiving end, the optical signal is fed into a photodetector where it is transformed into pulses of electric current which is then fed to decoder which converts the sequence of binary data stream into an analog signal which will be the

same information such as voice, which was there at the transmitting end.

Other Applications :

To mention a few other applications, the following could also be listed.

1. Sensing Device :

Optical fibers can be used as sensing devices wherein they are employed to sense parameters such as pressure, voltage or current, and the information is then fed to a processor from which the information is secured.

2. Data link :

A data link is a communication over a distance which is much shorter than what is required for a telecommunication trunk, but the required reliability is normally much higher. In such applications, the communication made using metal cables suffer from problems such as cross talk, impedance mismatching, etc., which reduce the reliability. All such problems are overcome by employing optical fibers as means of communication.

3. Local Area Networks :

Optical fibers provide more efficient communication facilities in local area networks wherein the information is required to be exchanged between terminals which are located at different places.

Numerical examples:

- **Example 1:** The refractive indices of core and cladding are 1.50 and 1.48 respectively in an optical fiber. Find the NA and angle of acceptance.

$$\begin{aligned}\text{Sol: } NA &= \sqrt{n_1^2 - n_2^2} \\ &= 0.244 \\ \text{Angle of acceptance} \\ \theta_a &= \sin^{-1}(NA) \\ &= 14.1^\circ\end{aligned}$$

Example 2: A step index fiber has a NA of 0.26, core refractive index of 1.5 and core diameter of 100 μm . Calculate the refractive index of the cladding.

$$\begin{aligned}\text{Sol: } NA &= \sqrt{n_1^2 - n_2^2} \\ \text{Substitute and square on both sides, we get} \\ n_2 &= 1.47\end{aligned}$$

- **Example 3:** The core refractive index of the optical fiber is 1.40, its relative refractive index is 2.5%. Determine the NA and the critical angle.

$$\begin{aligned}\text{Sol: } NA &= n_1 \sqrt{2} \Delta \\ \Delta &= 2.5\% = 0.025 \\ NA &= 0.313 \\ \text{The relative refractive index } \Delta &= \frac{n_1 - n_2}{n_1} \\ &= 1 - \frac{n_2}{n_1} \\ \frac{n_2}{n_1} &= 1 - \Delta \\ &= 0.975 \\ \text{The critical angle is } \theta_c &= \sin^{-1}\left(\frac{n_2}{n_1}\right) \\ &= 77.16^\circ\end{aligned}$$

- **Example 4:** Calculate the refractive indices of the core and cladding of a fiber from the following data, $NA=0.27$ and $\Delta=0.015$

$$\begin{aligned}\text{Sol: } NA &= n_1 \sqrt{2} \Delta \\ n_1 &= 1.558 \\ n_2 &= n_1(1 - \Delta) = 1.534\end{aligned}$$

5. Calculate the numerical aperture and hence the acceptance angle for an optical fiber whose core and cladding has refractive index of 1.45 and 1.40 respectively.

Solution:

If n_1 and n_2 are the refractive indices of core and cladding respectively, then, numerical aperture is given by,

$$\begin{aligned}N.A. &= \sin\theta_0 = \sqrt{n_1^2 - n_2^2} \\ &= (1.45^2 - 1.40^2)^{1/2} \\ &= 0.38 \text{ (Ans.)}\end{aligned}$$

$$\text{Acceptance angle } \theta_0 = \sin^{-1}(0.38) = 22.3^\circ \text{ (Ans.)}$$

Review Questions

1. What is luminescence? Mention the various methods by which luminescence can be achieved. (4 marks)
2. What is a LASER? Explain its properties. (4 marks)
3. Distinguish between spontaneous emission and stimulated emission. (4 marks)
4. Explain how Einstein's theory predicts existence of stimulated emission. (4 marks)
5. Define the terms 'Metastable state' and 'Population inversion'. (4 marks)
6. Mention the basic conditions for laser action. How can they be materialized? (4 marks)
7. What are the basic components required for the construction of a laser? Explain the method of production of laser. (8 marks)
8. Discuss the relative advantages and disadvantages of three level and four level lasers. (8 marks)
9. What are the different methods of achieving population inversion? Explain. (4 marks)
10. Describe the construction and working of a Nd:YAG laser. What are its disadvantages? (8 marks)
11. Describe the construction and working of a semiconductor laser. (8 marks)
12. Briefly explain the applications of lasers. (4 marks)
13. Explain the applications of LASER in bar code scanner. (4 marks)
14. Explain how LASER is used in printer. (4 marks)
15. What is an optical fiber? What is the principle on the basis of which optical transmission is achieved through a fiber? Explain. (4 marks)
16. Explain the terms numerical aperture, semi angle of the cone of acceptance and relative refractive index difference. (8 marks)
17. Derive an expression for numerical aperture of an optical fiber in terms of R.I. of core and cladding. (8 marks)
18. What are the different modes of propagation possible in optical fibers? Explain in detail with necessary diagrams. (8 marks)
19. Write any four differences between single mode and multi-mode fibers. (4 marks)
20. What is attenuation in an optical fiber? Mention the possible reasons for the same. (4 marks)
21. Describe how the optical fiber can be used in communication. Mention its advantages. (4 marks)
22. Describe an optical fiber with working principle. Write any two advantages of an optical fiber communication system over conventional communication system. (8 marks)
23. Describe the different types of optical fiber along with the sketches of geometry, refractive index profile and ray propagation. (8 marks)

Related Problems: (4 marks)

24. Find the ratio of population of the two energy states the transition between which is responsible for the emission of photons of wavelength 694.3 nm. Assume the ambient temperature as 27°C.
25. The ratio of population of two energy level is 1.059×10^{-30} . Calculate the wavelength of the emitted photon at 300 K.

26. The ratio of population of two energy level is 1.08×10^{-30} . Calculate the temperature, if the wavelength of the emitted photon is 696 nm.
27. A laser beam has pulse duration of 12ps and a power of 200mW. Calculate the number of photons in each pulse, if the laser wavelength is 1060 nm.
28. A He-Ne laser emits light at a wavelength of 632.8 nm and has an output power of 2.3 mW. How many photons are emitted in each minute by this laser when operating?
29. Calculate the wavelength of emission from a GaAs semiconductor laser whose band gap energy is 1.44 eV.
30. The angle of acceptance of an optical fiber is 30° when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33
31. A fiber has a core diameter of 6 μm and its core refractive index is 1.47 and for cladding it is 1.43. How many modes can propagate into the fiber if the wavelength of the source is 1.5 μm .
32. The fractional index change of optical fiber and refractive index of core are 0.0515 and 1.533 respectively. Determine the refractive index of cladding.
33. The fractional refractive index of an optical fiber of diameter 80 μm is 0.03. The refractive index of the core is 1.48 and the operating wavelength is 0.8 μm . Calculate (a) acceptance angle (b) numerical aperture, (c) critical angle for core-cladding interface and (d) number of modes supported in the fiber.
34. In multimode optical fiber with core R.I. 1.49 and cladding R.I. 1.48, can support 1200 modes. If the wavelength of light is 1.25 μm determine the core diameter.
35. Calculate the acceptance angle and critical angle for the core-cladding interface when the core R.I. is 1.48 and fractional index change is 2%.
36. A step index fiber has a numerical aperture of 0.26, a core refractive index of 1.5 and a core diameter of 100 μm . Calculate the R. I. of cladding, angle of acceptance and the maximum number of modes with a wavelength of 1 μm that the fiber can carry.
37. A glass clad fiber is made with core glass of refractive index 1.5 and the cladding is doped to give a fractional index difference of 0.05. Determine (a) the acceptance angle (b) the numerical aperture and (c) the critical angle (also known as critical internal reflection angle).
38. The refractive index of the core and cladding are 1.48 and 1.45 respectively. Calculate the acceptance angle of the fiber. If the diameter of the fiber is 60 μm , find the number of modes the fiber can support at an operating wavelength of 1.5 μm .
39. A laser operating at 632.8 nm emits 3.18×10^{19} photons per second. Calculate the output power of the laser if the input power is 100 watts. Also find the percentage power converted into coherent light energy.
40. A Nd:YAG laser emits pulse of power of 1W and of duration 12ps . Calculate the number of photons in each pulse, if the laser wavelength is 1064nm.
41. An optical fiber has a core material with refractive index 1.55 and its cladding material has a refractive index of 1.50. The light is launched from air. Calculate its numerical aperture, the acceptance angle and also the fractional index change.

42. Calculate the V-number for a fiber of core diameter $40\text{ }\mu\text{m}$ and with refractive indices of 1.55 and 1.50 respectively for core and cladding when the wavelength of the propagating wave is 1400 nm . Also calculate the number of modes that the fiber can support for propagation. Assume that the fiber is in air.
43. Find the attenuation in an optical fiber of length 500 m , when a light signal of power 100 mW emerges out of the fiber with a power 90 mW .
44. Calculate the refractive indices of core and cladding materials of an optical fiber if its numerical aperture is 0.22 and relative refractive index difference is 0.012 .
45. The angle of acceptance of an optical fiber is 30° when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33
